

Minoan names built on plant names? Five candidates from Linear A, and the measurement of their criterion

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Summary

Greek formed place names and personal names on plant names, and several of those plant names are themselves pre-Greek. This note asks whether the names of the Minoan archives do the same, and measures the criterion it uses to answer.

Three sign groups are examined: **KU-PA** against *kyparos* / *ku-pa-ro* (galingale), **KU-MI-NA-QE** against *ku-mi-no* (cumin), and **KI-KI-RA-JA** beside Neumann's KI-KI-NA (sycomore fig). Each is established first by distribution — position in the document, sealing group, attested derivatives, co-occurring commodity logogram — and only then compared in form, since a method that starts from sound has no discriminating power in this corpus: read with Linear B values and reduced to consonant skeletons, 47% of Linear A groups are attested in Mycenaean Greek, 43% in Hittite, 62% in Ugaritic, 70% in Hebrew and 50% in Etruscan, and the highest ratio belongs to the largest lexicon.

The criterion these three readings rest on has now been measured, and it does not reach significance. Of the 661 intact groups in the corpus, 269 (41%) satisfy the distributional condition, 504 (76%) the derivative-or-logogram condition, and 165 (25%) both. Adding root agreement with a list of twenty plant names, **five groups satisfy all three:** KU-PA, KU-MI-NA-QE, KI-KI-RA-JA, KA-NA and KI-TA-I. Against a null of false plant names built from real syllables of the corpus, the mean is 1.74 and the maximum 9 in two thousand permutations, giving **p = 0.0515**.

Two consequences follow, and they govern this version. The three readings are presented as **candidates with their measurement declared**, not as identifications. And **KA-NA and KI-TA-I, which satisfy exactly the same criterion, are added to the list:** selecting three of five without a prior rule would be the kind of post-hoc choice this work measures in other proposals.

Keywords: Linear A, Minoan, onomastics, phytonyms, null models, Hagia Triada.

1. The question, and why it has not been asked

That Greek formed toponyms and anthroponyms on plant names is well known: Kyparissos and Kyparissia (κυπάρισσος, cypress), Marathon (μάραθον, fennel), Selinous (σέλινον, celery), Rhodos (ρόδον, rose), Kydonia (the quince), and among personal names Hyakinthos, Narkissos, Daphne. Several of those plant names are themselves pre-Greek (Beekes 2010; Furnée 1972), and some appear already in Mycenaean as toponyms (*ku-pa-ri-so* at Pylos) or as commodities (*ku-pa-ro*, *ku-mi-no*, *sa-sa-ma*, *ko-ri-ja-do-no* in the Ga series at Knossos, Ge at Mycenae and Un at Pylos).

For Linear A the only accepted identification of a plant word is Neumann’s (1960): KI-KI-NA in HT 88 as the sycomore fig (Greek κίκι, Egyptian *kḳḳ*), supported by the group following the fig logogram (NI · KI-KI-NA 7). Younger (2024a) records dozens of groups in his lexicon as “name in a list”, “heading” or “place name?” without proposing etymologies. Jiménez Delgado (2024) has collected the pre-Greek names in -to of Linear B and compared them with Linear A sign groups. **The question of whether the names of the Minoan archives contain plant words has not been posed as such.**

2. Why the form cannot come first

The interpretation of Linear A sign groups rests on a small number of secure identifications obtained from internal analysis: KU-RO “total” and KI-RO “deficit” (Duhoux 2011), a set of transaction terms and headings (SA-RA₂, KA-PA, A-DU; Schoep 2002; Montecchi 2010), and the toponyms and anthroponyms that reappear in the Linear B of Knossos (Godart 1984). Identifications proposed on phonetic grounds alone have not gained acceptance, for the reasons set out by Duhoux (1989).

That observation can be quantified. For the 729 Linear A sign groups with at least two consonants, read with the values of their Linear B homographs and reduced to a consonant skeleton over a common inventory of thirteen classes, the proportion of skeletons attested in a comparison lexicon is 47% for Mycenaean Greek, 43% for Hittite, 62% for Ugaritic, 70% for Hebrew and 50% for Etruscan; against each lexicon with its consonants permuted, the ratios are 1.47, 1.46, 1.50, 1.68 and 1.57. **No comparison language stands out, and the highest ratio belongs to the largest lexicon.**

The procedure followed here is therefore the one that produced KU-RO: the function of a sign group is established from its behaviour in the documents, and external comparison is admitted afterwards, as corroboration. Two additions are made. Each distributional claim is tested against a permutation null; and the same tests were calibrated on Etruscan, where they recover the genitive before *clan* (58% of words against a base rate of 26%) and the northern distribution of the marked sibilant.

3. How many groups would satisfy the three conditions by chance

For a match to count, the first version of this work required three conditions: **root agreement** in the first two syllables with an attested plant name; **the distribution of a party** (heading a document, or appearing on a roundel of the sealing group that pairs sign group with commodity logogram); and **an attested derivative or a coherent co-occurring logogram**.

An argument that rests on individual cases satisfying three conditions must measure what chance produces under those same conditions. That test was missing.

Over the 661 intact groups of the corpus:

condition	groups	share
distribution of a party	269	41%

condition	groups	share
derivative or co-occurring logogram	504	76%
both	165	25%
all three , with a list of twenty plant names	5	0.8%

The five are **KU-PA, KU-MI-NA-QE, KI-KI-RA-JA, KA-NA and KI-TA-I**.

Against a null of false plant names built from real syllables of the corpus, preserving the number of plant names and the length of the root, the mean is **1.74** and the maximum **9** in two thousand permutations. The observed value of five gives **p = 0.0515**.

The three conditions do not reach significance. The excess is nearly threefold and the three groups proposed in the first version are among the five the criterion selects, but a criterion whose p stands at 0.0515 does not license an identification.

An earlier control pointed the same way and is repeated here. Screened against two plant-name lists, one of sixty and one of seventy-nine, the full vocabulary of 799 groups gives 91 root matches against 53.1 expected with the first list, but 97 against 73.7 with the second: **the excess of the shorter list concentrates in the KU-PA family and disappears when the list is broadened.** The same screen applied to the Linear B corpus (5,597 groups) produces 383 matches, most with Greek anthroponyms unrelated to plants (*ke-re-te-u, ko-re-te*), confirming that two-syllable agreement alone carries no weight.

4. The five candidates

KU-PA. Against *kyparos / ku-pa-ro*, galingale. Attested with the derivatives KU-PA-JA and KA-KU-PA, and on roundels of the second Haghia Triada sealing group, where KA-KU-PA is stamped by CMS II,6 n° 11 with logogram *303 (three impressions) and by n° 21 with *629 and *624. In that group the sign group is the party and the logogram the commodity, since one seal accompanies eight different groups. Younger (2024b) reads KA-KU-PA at HT 16 as the person responsible for the contributions of *303, which is compatible.

KU-MI-NA-QE. Against *ku-mi-no*, cumin. On roundel HT Wc 3014 with the goat logogram, stamped by CMS II,6 n° 77. **A qualification is due here to an observation by J. M. Jiménez Delgado (University of Seville):** the final -QE is unexplained. Seven groups of 782 types end in -qe (0.9%), only two have their root attested separately, and against a null of shuffled syllables those two pairs give **p = 0.086**, so -qe does not hold as a productive suffix, unlike -ja, which does with seven pairs (p = 0.020). And KA-PA-QE is telling: its root KA-PA is a Cretan toponym (Uchitel 2002-2003) heading six Haghia Triada tablets, so KA-PA-QE reads naturally as “and KA-PA”. If Minoan had an enclitic copula comparable to Mycenaean -qe, KU-MI-NA-QE would be the word *ku-mi-na* with a particle attached and not a name. **The case satisfies two of the three conditions and the third is open.**

KI-KI-RA-JA. Beside Neumann's KI-KI-NA, the sycomore fig. Heads HT 85b, which Uchitel and Finkelberg (1995) had already identified as a heading.

KA-NA and KI-TA-I satisfy the same three conditions and were not proposed in the first version. They are listed here for the reason given in the summary: a criterion selects what it selects.

5. What this note claims, and what it does not

It claims that the question is worth posing and that the criterion for answering it can be measured. It does not claim that any of the five groups is a plant name.

The distinction matters because the field's difficulty with Linear A is not a shortage of proposals but the absence of a stated criterion by which one proposal can be preferred to another. **A criterion at $p = 0.0515$, declared, is more useful than three readings asserted without one.**

Statement of artificial intelligence assistance

The measurements and the drafting of this paper were carried out with the assistance of a language model (Claude, Anthropic), under the design, verification and responsibility of the author.